LASC MED201

Class-1&2

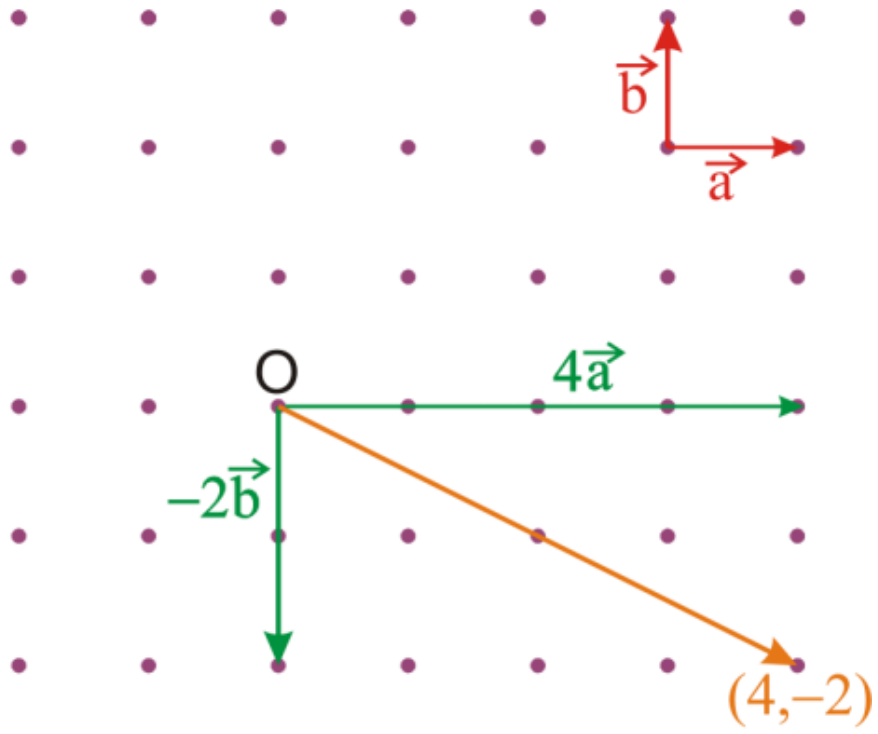
Miller Notation

- Miller Indices (for cubic system)
  - Direction
  - Planes
- Miller-Bravais (for Hexagonal system)

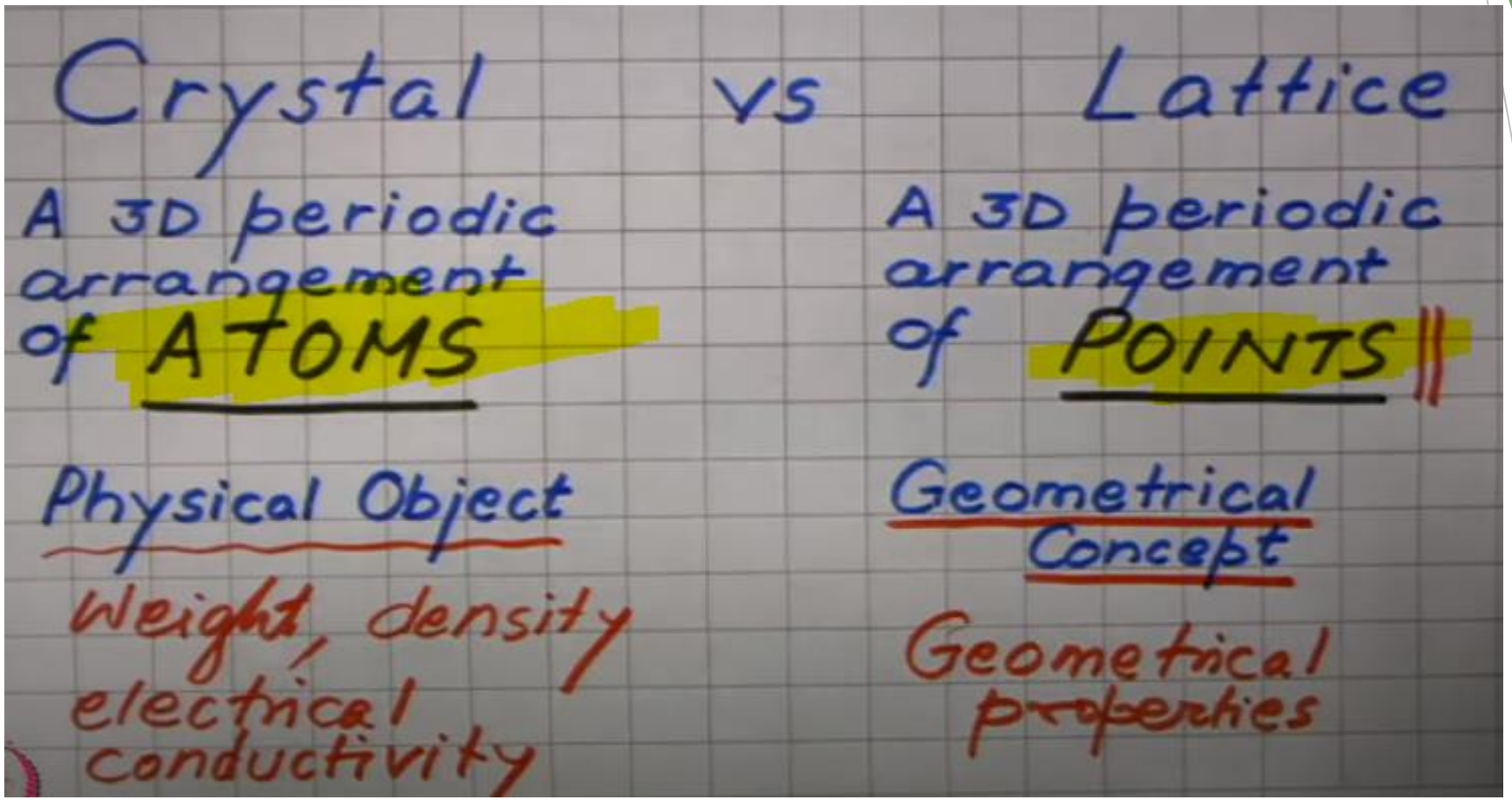
Miller Indices

- Miller indices are used to specify directions and planes
- The number of indices will match with the dimension of the lattice or the crystal.
- E.g. in 1D there will be 1 index and 2D there will be two indices etc
- Negative numbers/directions are denoted with a bar on top of the number

Example



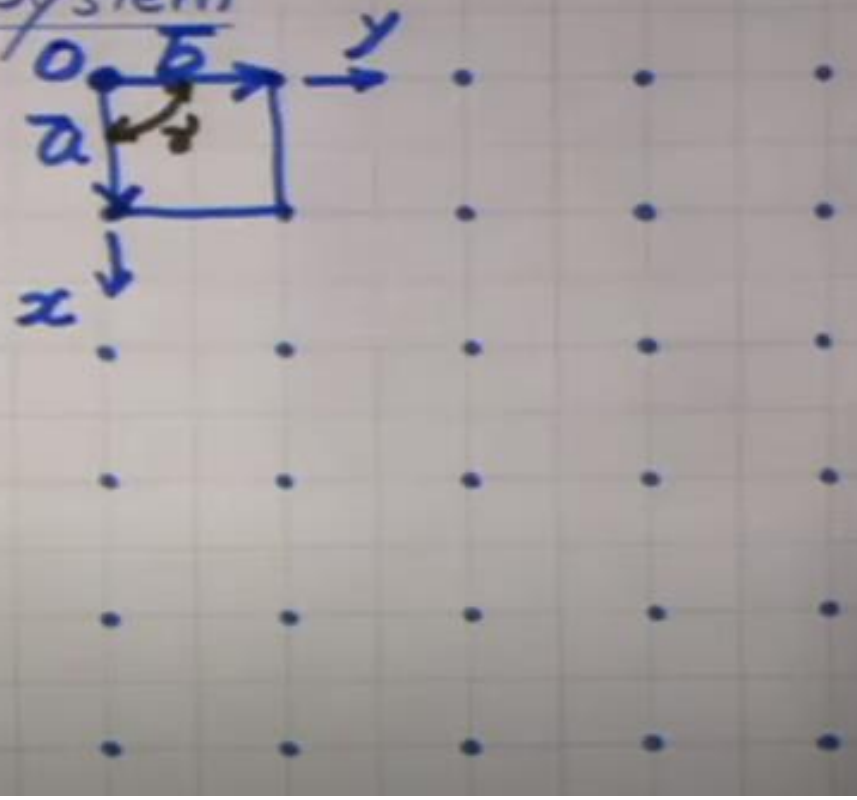
$$Miller\ Indices \rightarrow [4\bar{2}]$$



Source: IMSE:YT

Crystal Coordinate System

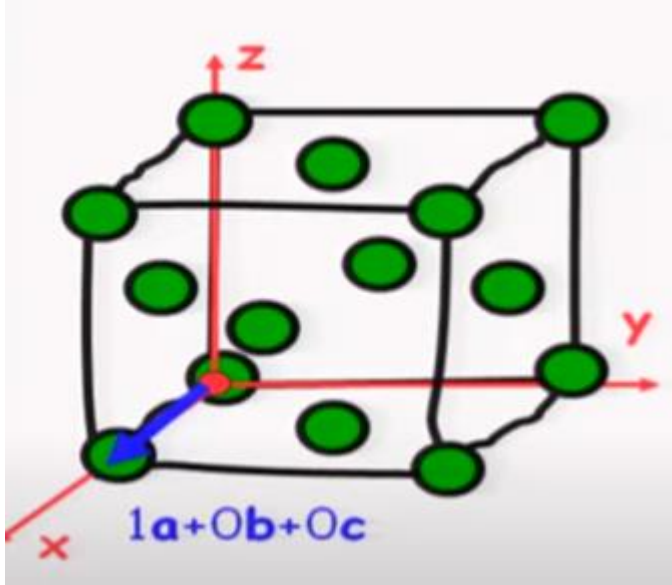
Select a corner of the unit cell as the origin. The vectors from the corner along the edges of the unit cell define basis vector of the lattice.



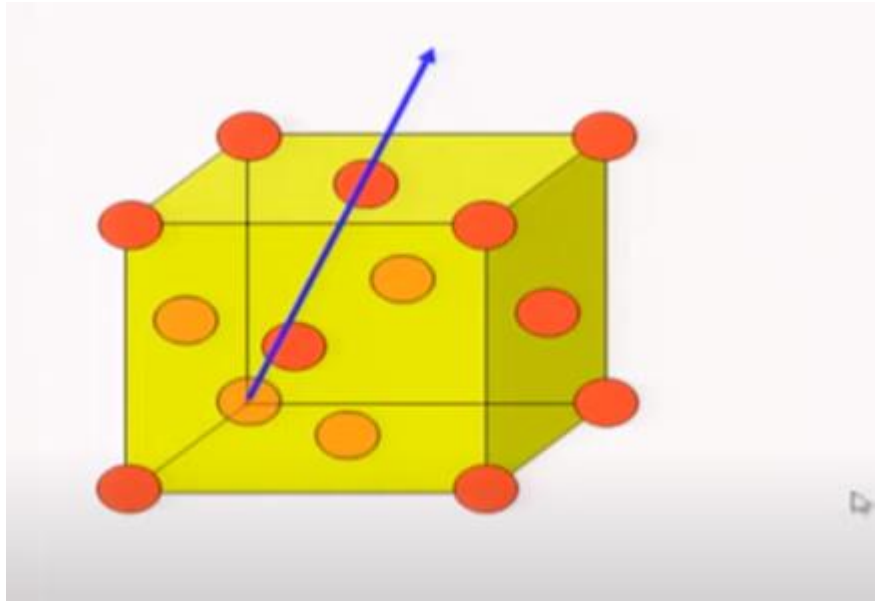
Miller Indices: Direction [uvw]

- Rules to be followed

1. Choose a <u>point on direction as origin</u>
2. Choose a coordinate system with axes parallel to the unit cell edges
3. Find the coordinates of another point on the direction in terms of a,b,c (lattice parameters)
4. Reduce the integer to smallest integer.
5. Put in square bracket. [100]

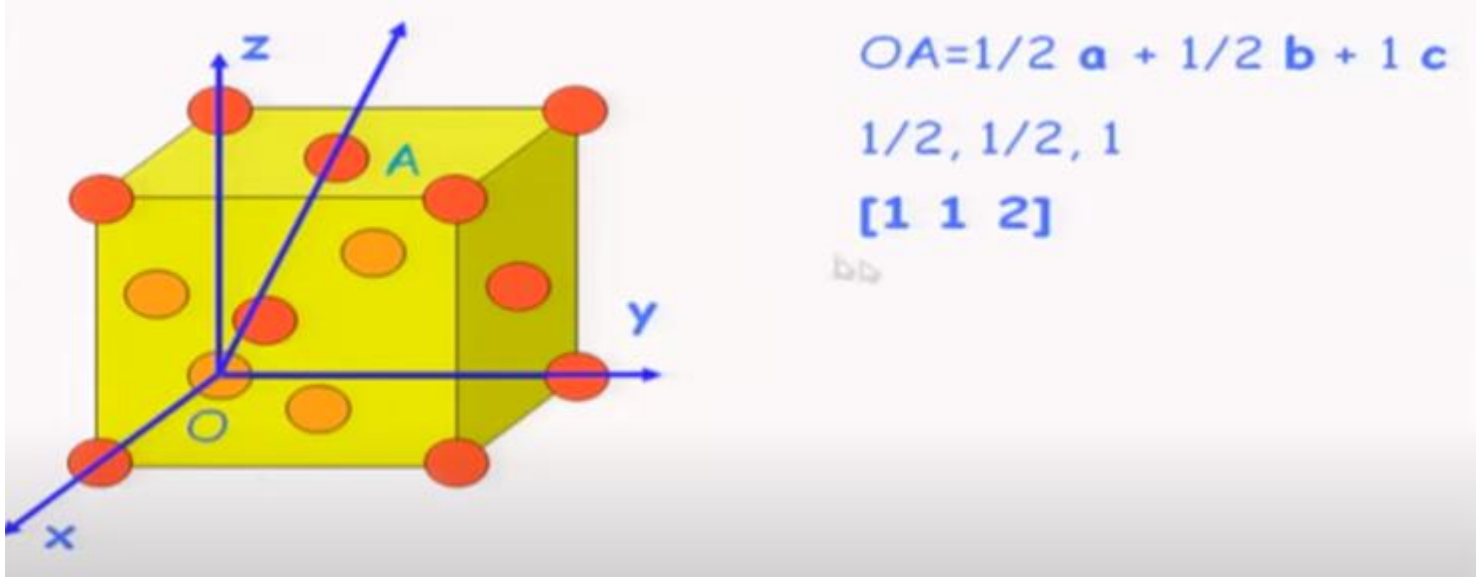


Example-1

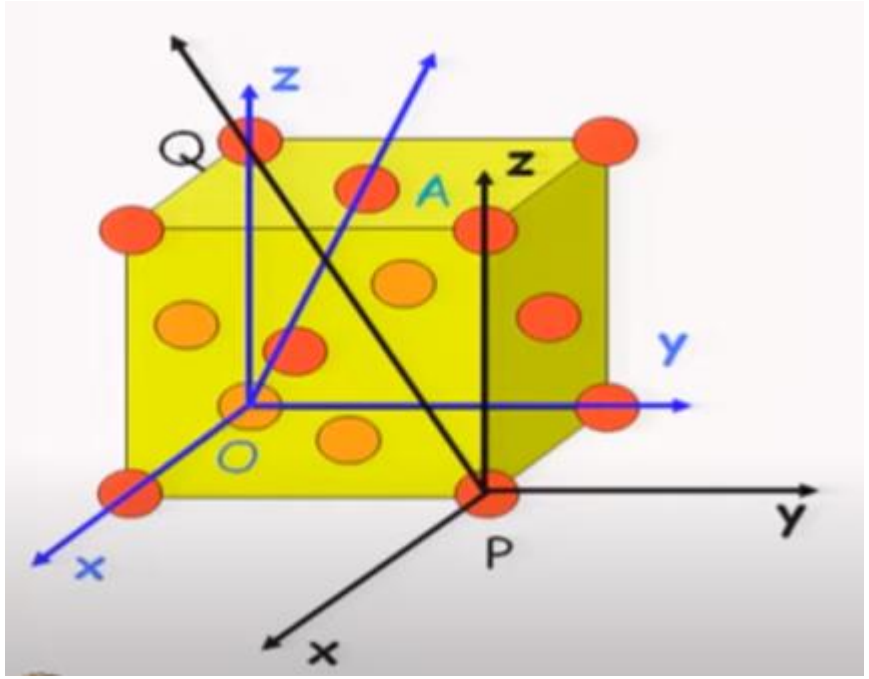


Find miller indices for the direction in blue

Solution-1



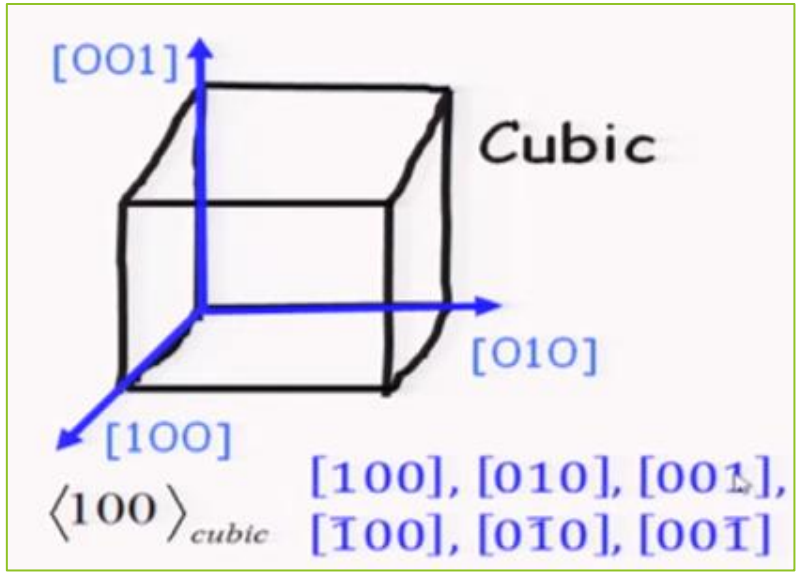
Example-2

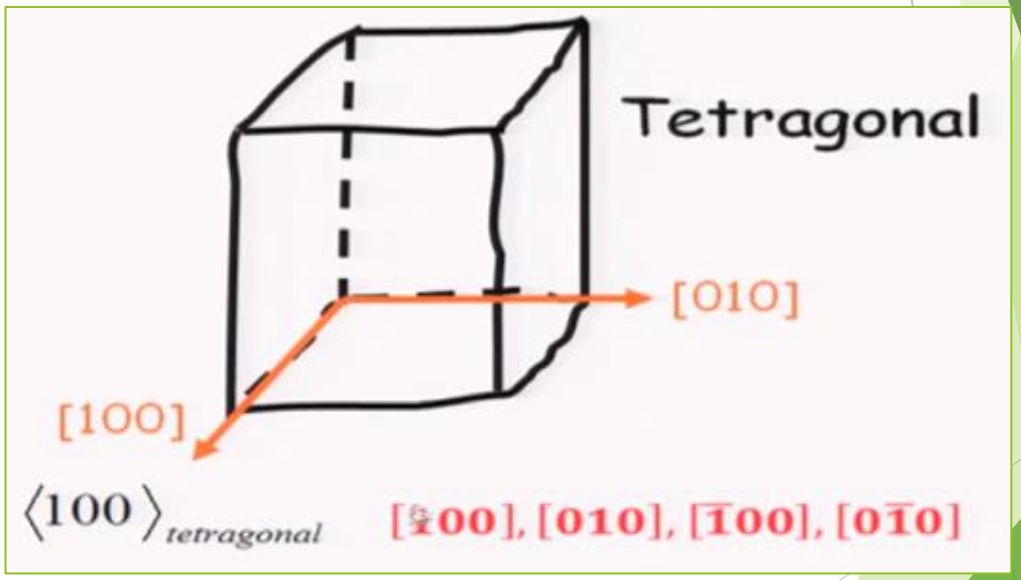


[1bar 1bar 1]

Family of Directions <uvw>

- <uvw> : [uvw] and all other directions related to [uvw] by the <u>symmetry of crystal</u>





Miller Indices: Planes (hkl)

- Rules to be followed:

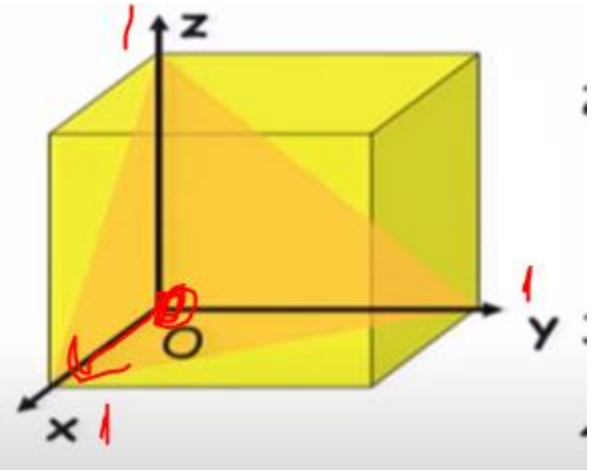
1. Select a crystallographic coordinate system with **origin not on the plane.**

2. Find the intercepts along axes in terms of respective lattice parameters. If intercept is parallel to axis then consider it to be infinity.

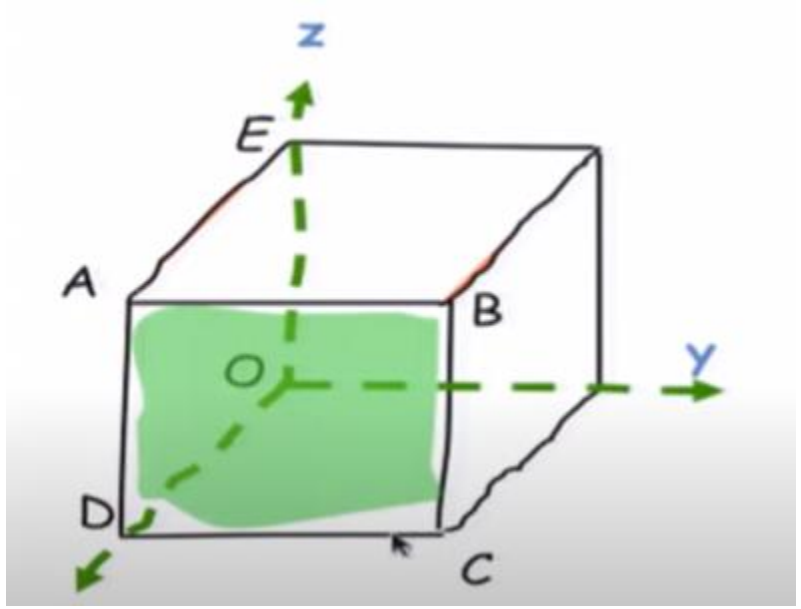
3. Take reciprocal

4. Convert to smallest interger

5. Enclose in paranthesis (111)



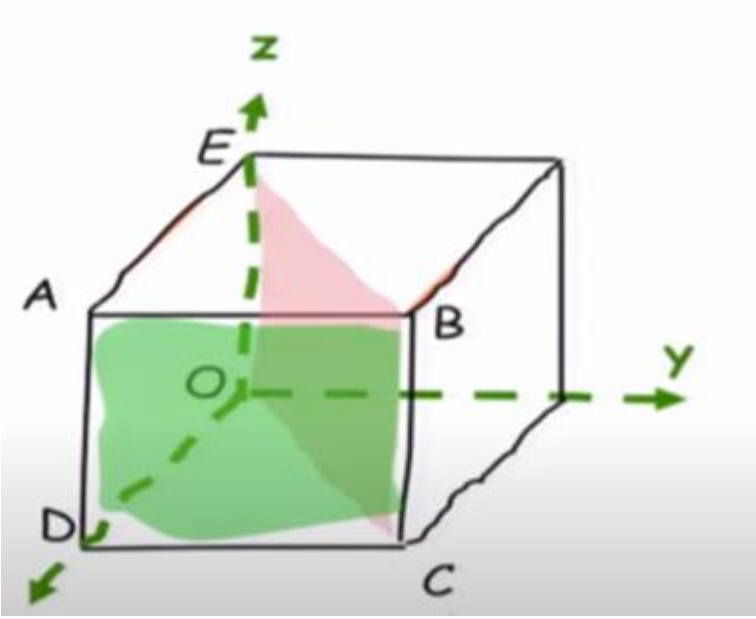
Example-1



Plane	ABCD
origin	O
intercepts	1 ∞ ∞
reciprocals	1 0 0
Small integers	1 0 0
Miller Indices	(1 0 0)

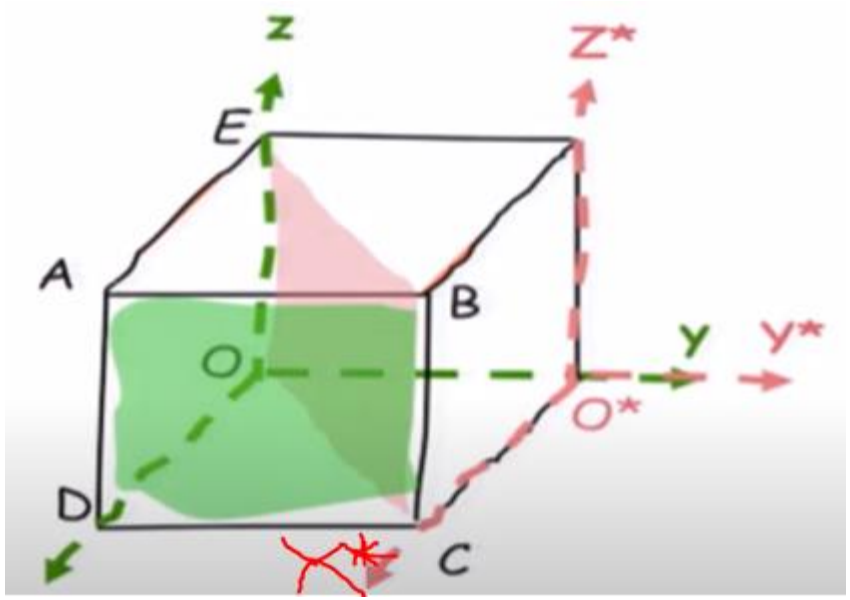
Inference: 0 here represents that the plane is parallel to that corresponding axis

Example-2



Find miller indices for plane highlighted in pink.

Solution-2

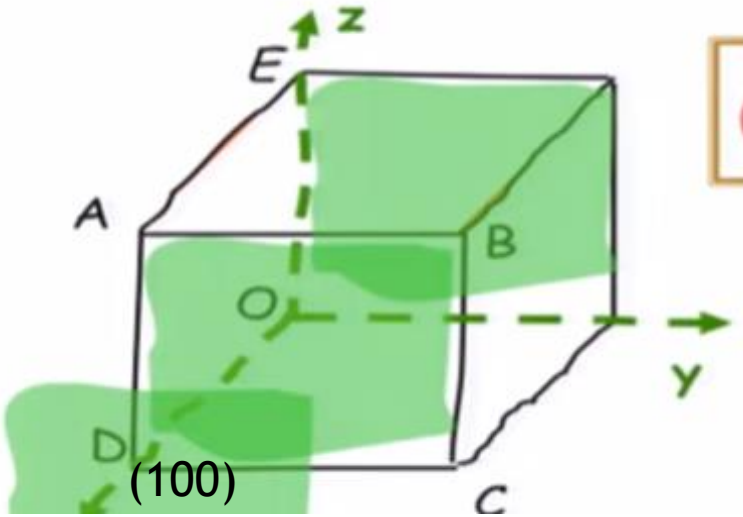


Plane	OCBE
origin	O*
intercepts	1 -1 ∞
reciprocals	1 -1 0
Miller Indices	$(1\ \bar{1}\ 0)$

Important Point

Miller indices of a plane specifies only its orientation in space not its position

All parallel planes have the same Miller Indices



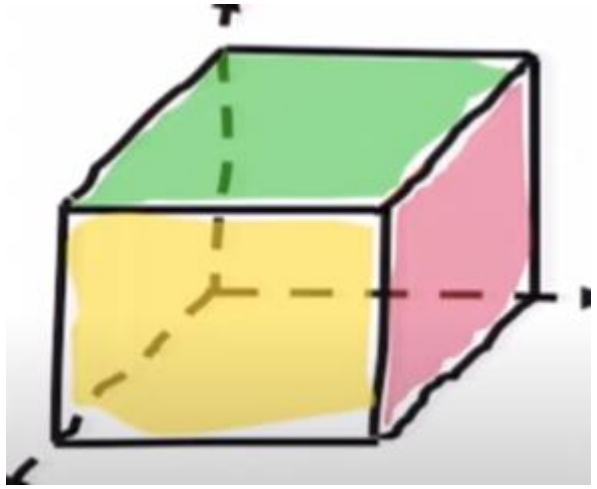
$$(h\ k\ l) = (\bar{h}\ \bar{k}\ \bar{l})$$

$$(100) = (\bar{1}00)$$

Source: IMSE:YT

Family of Planes {hkl}

- {hkl}: (hkl) and all the other planes related to (hkl) by the symmetry of the crystal.

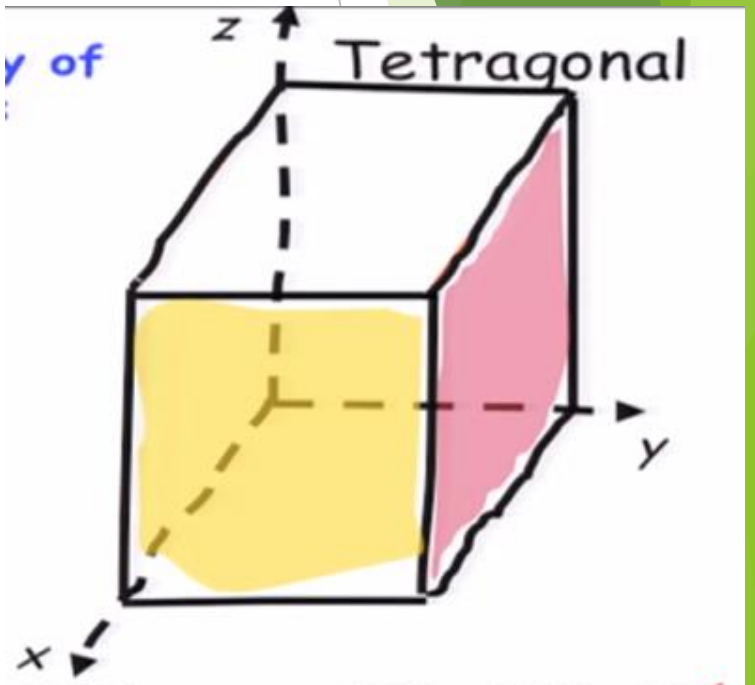


Front & back faces: (100), $(\bar{1}00)$

Left and right faces: (010), $(0\bar{1}0)$

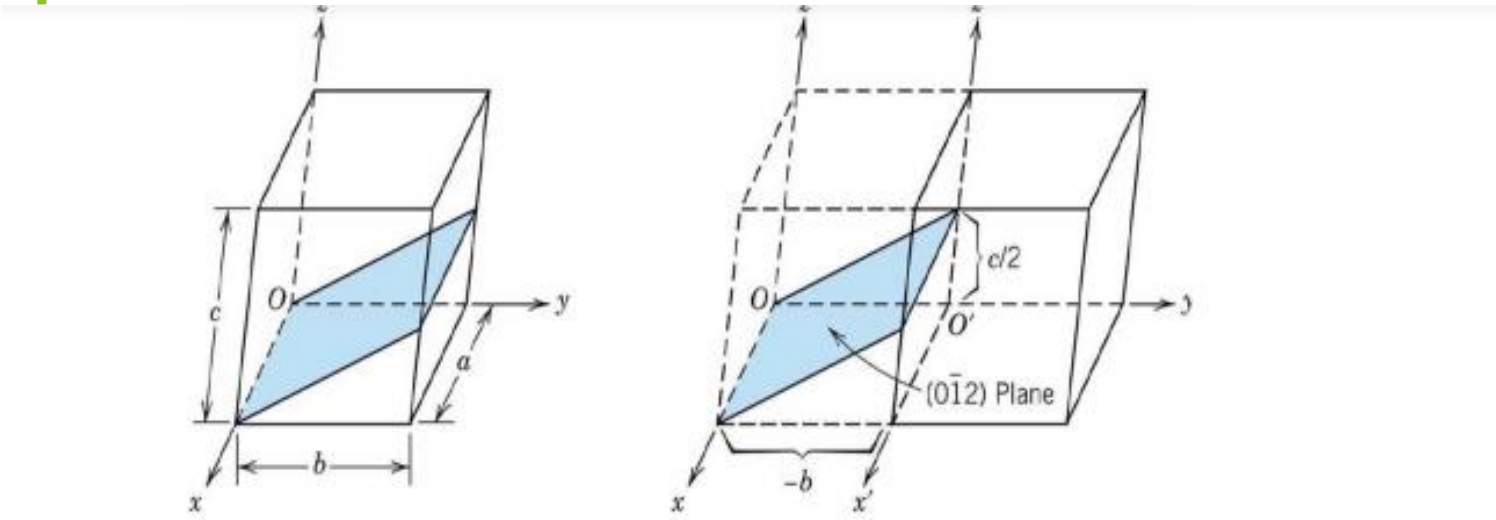
Top and bottom faces: (001) $(00\bar{1})$

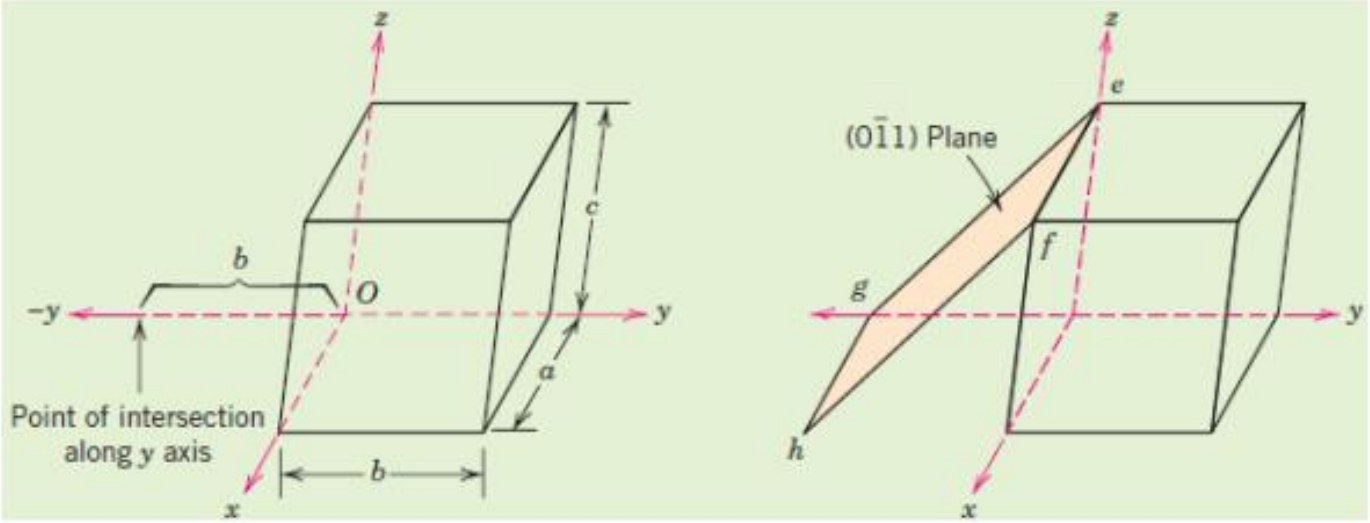
$$\{100\}_{cubic} = (100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$$



$$\{100\}_{tetragonal} = (100), (010), \cancel{(001)}$$
$$(\bar{1}00), (0\bar{1}0), \cancel{(00\bar{1})}$$

Example from Slides





Summarizing notations

- [uvw]: Miller indices of a direction (or set of parallel lines)
- (hkl): Miller indices of a plane (or set or parallel planes)
- <uvw>: Miller indices of a family of symmetry related directions
- {hkl}: Miller indices of a family of symmetry related planes

Useful Conventions for cubic crystals

- [*uvw*] is normal /perpendicular to (*hkl*) →
  $u = h$, $v = k$, and $w = l$. E.g.:(111) ⊥ [111], {100} ⊥ <100>

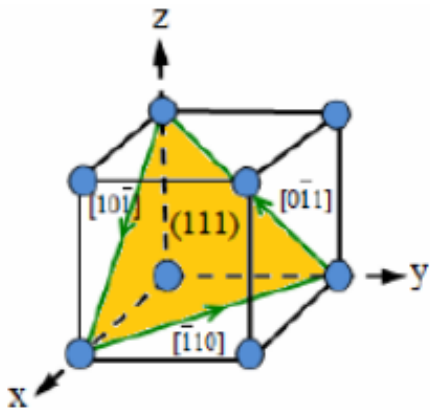
- [*uvw*] is parallel to (*hkl*) if $hu + kv + lw = 0$, [$\bar{1}$10] lies in (111)

- Two planes $(h_1k_1l_1)$ and $(h_2k_2l_2)$ are normal if $h_1h_2 + k_1k_2 + l_1l_2 = 0$

- Two directions $(u_1v_1w_1)$ and $(u_2v_2w_2)$ are normal if
  $u_1u_2 + v_1v_2 + w_1w_2 = 0$

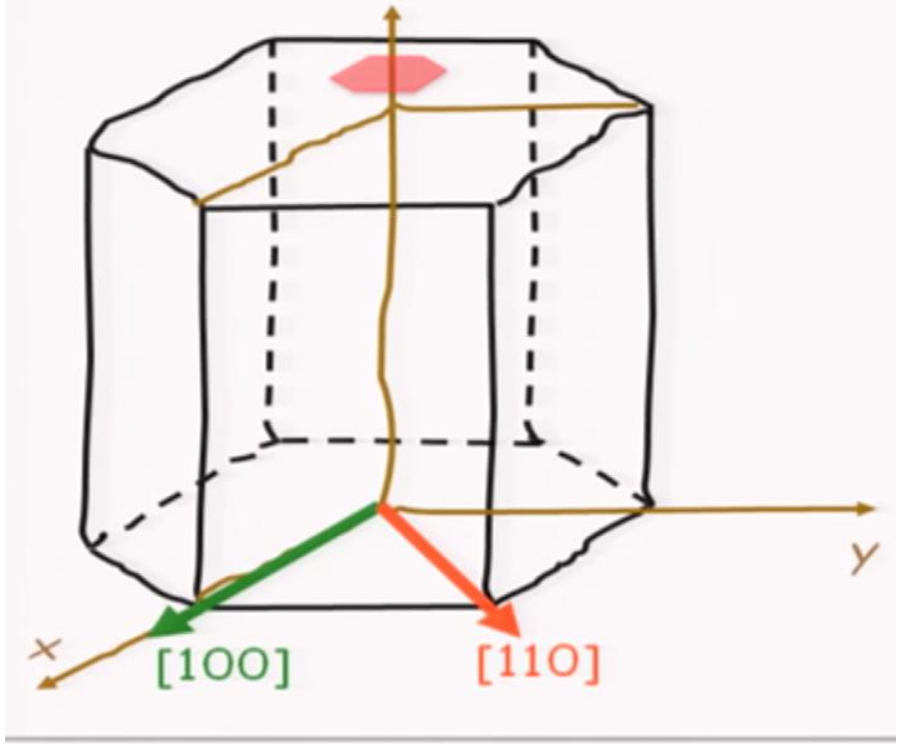
Angle between two planes is given by:

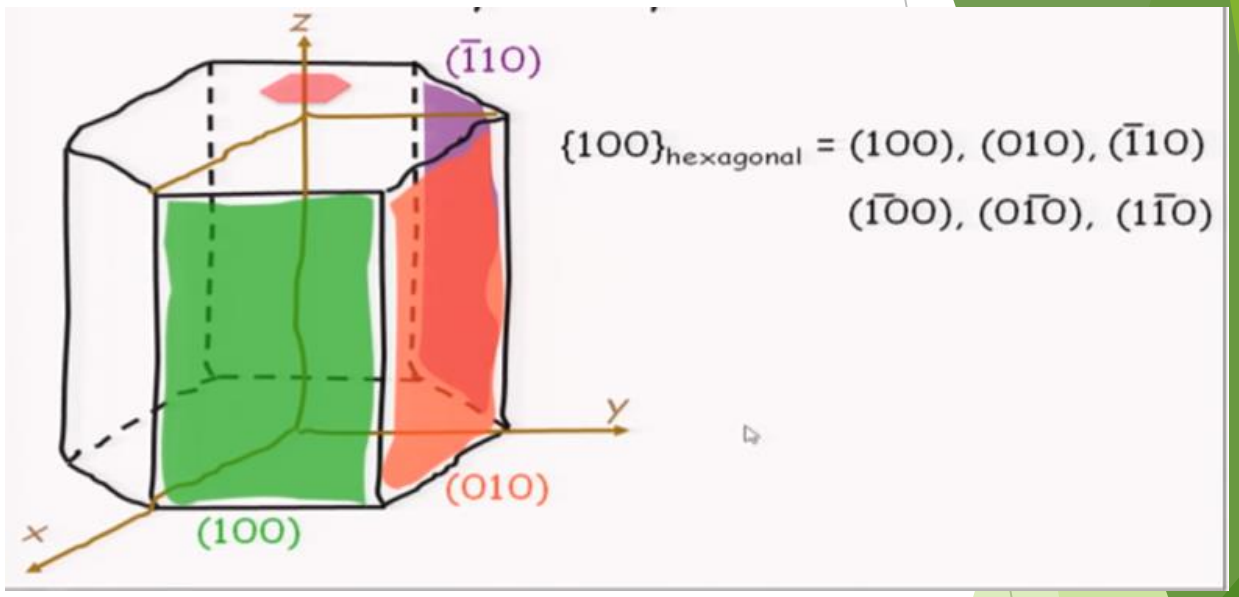
$$Cos\theta = \frac{h_1h_2 + k_1k_2 + l_1l_2}{\sqrt{h_1^2 + k_1^2 + l_1^2}\sqrt{h_2^2 + k_2^2 + l_2^2}}$$



Source: Slides by Prof Grewal

Miller-Bravais (4-index system)





$$\langle 100 \rangle_{hexagonal} = [100],\ [010],\ [110]$$
$$[\bar{1}00],\ [0\bar{1}0],\ [\bar{1}\bar{1}0]$$

Symmetry related directions or faces/planes should be permutations but here they are not

Need for 4th Index

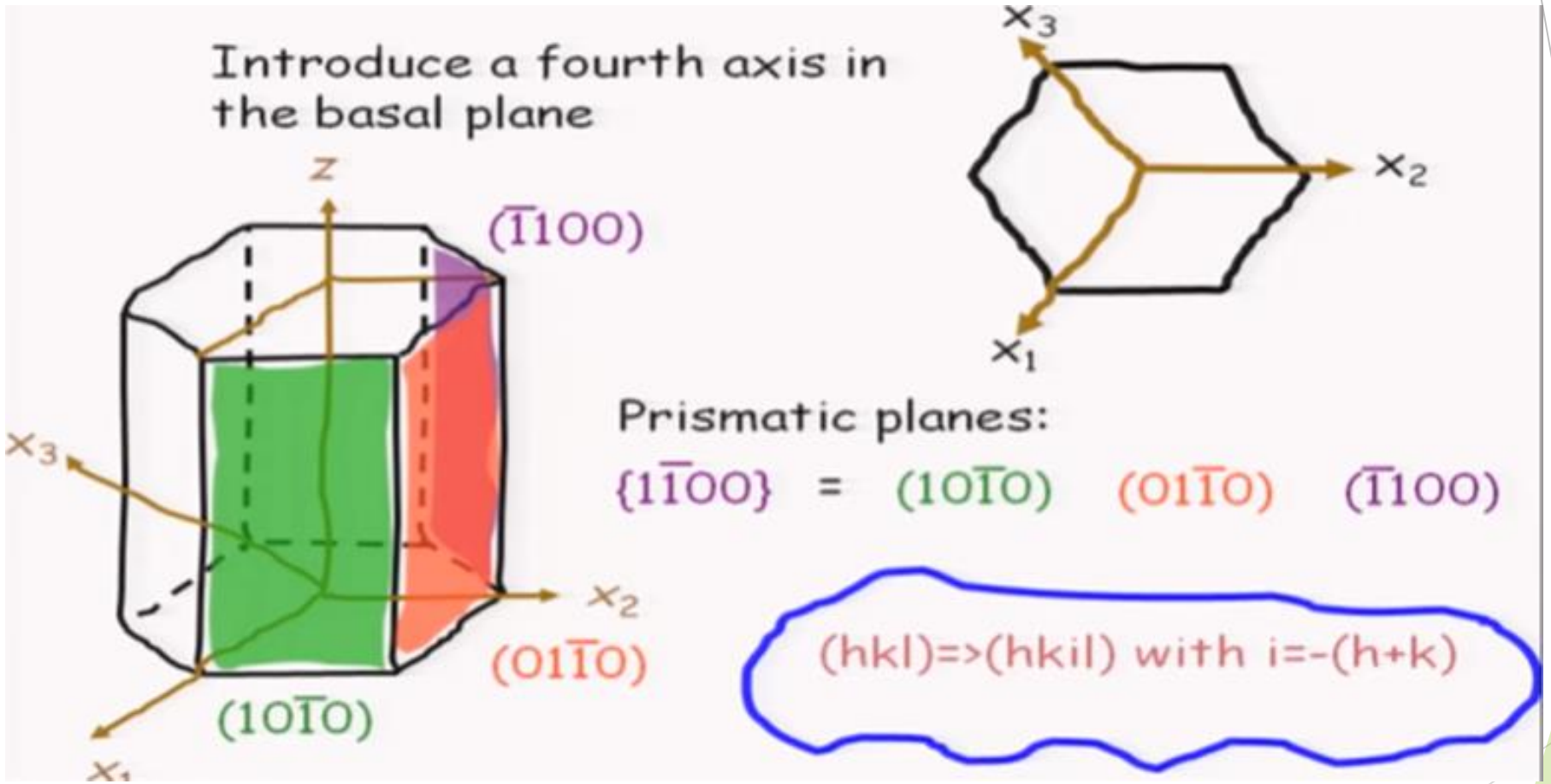
Problem:

In hexagonal system symmetry related planes and directions do *NOT* have Miller indices which are *permutations*

Solution:

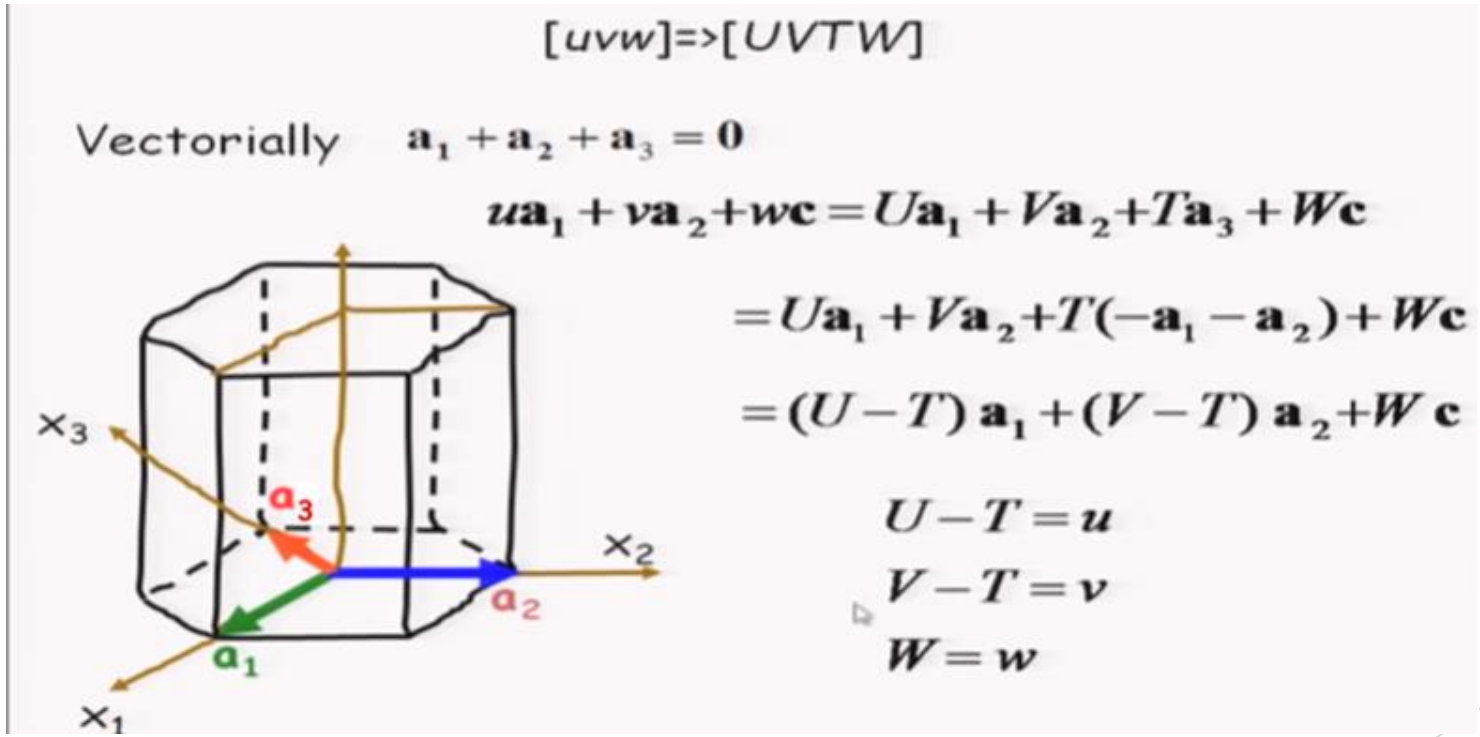
Use the four-index Miller-Bravais Indices instead

Miller-Bravais for plane (hkil)



Miller-Bravais for Direction [UVTW]

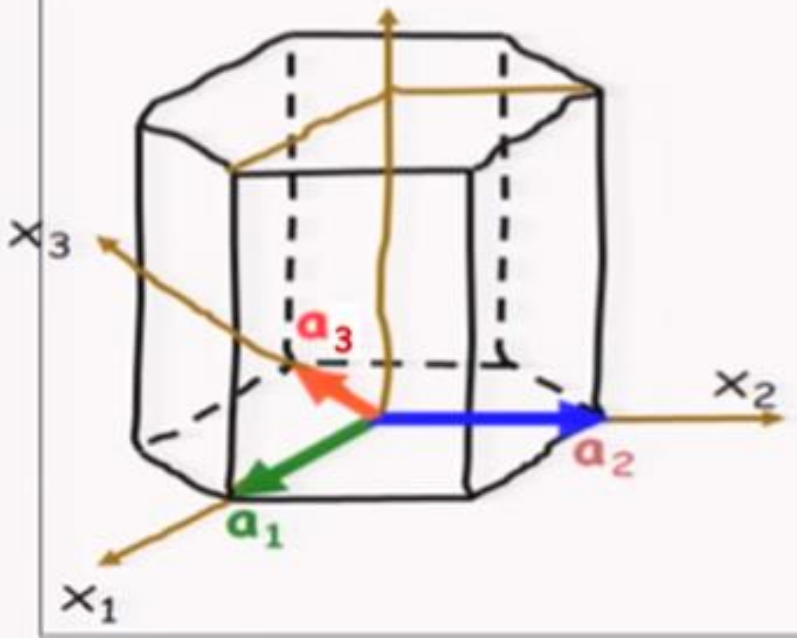
- Derivation: You can skip it totally. Putting it here just for reference of the final formula which is to be used.



$$U - T = u$$

$$V - T = v$$

Further require that: $U + V + T = 0$

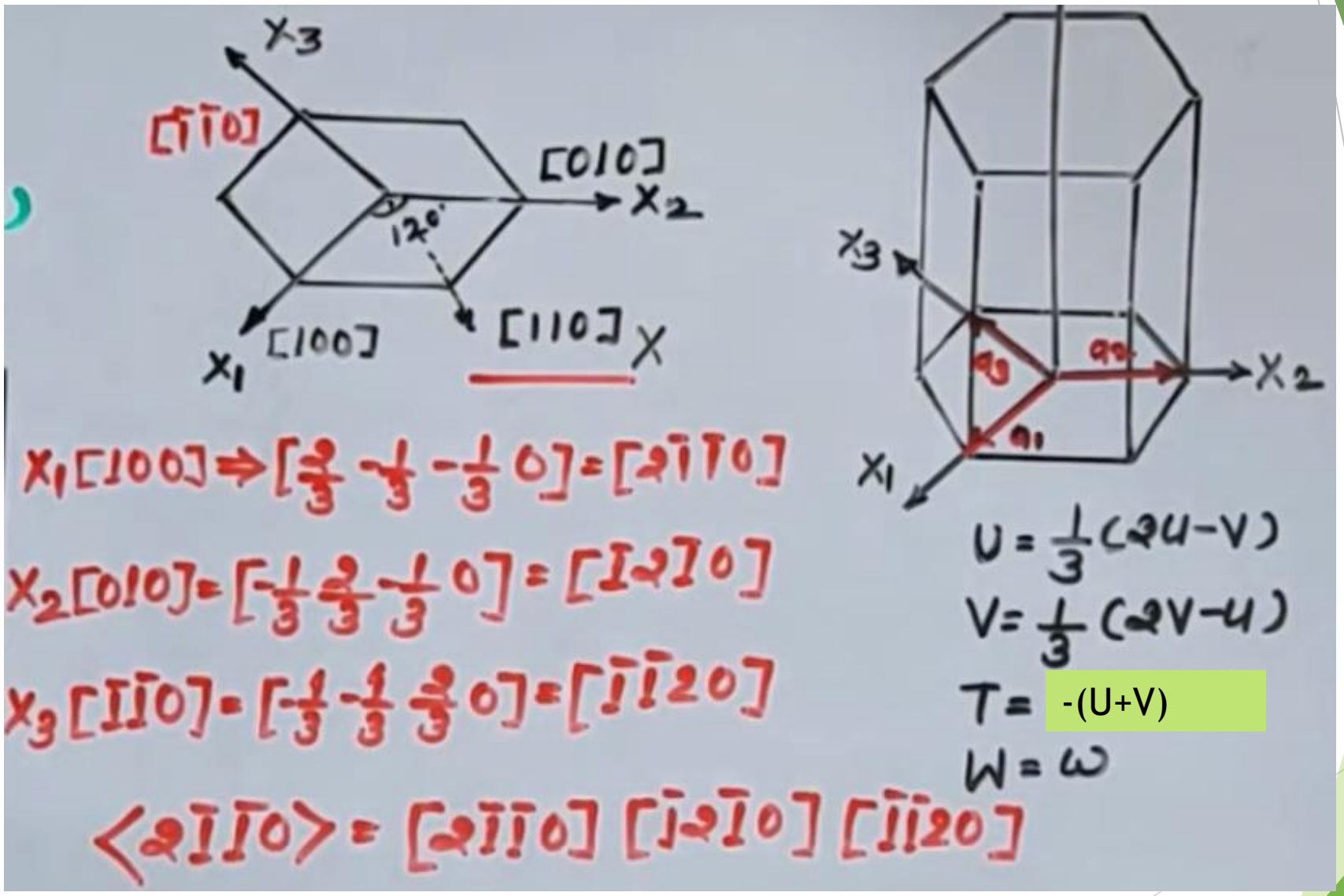


$$U = \frac{1}{3}(2u - v)$$

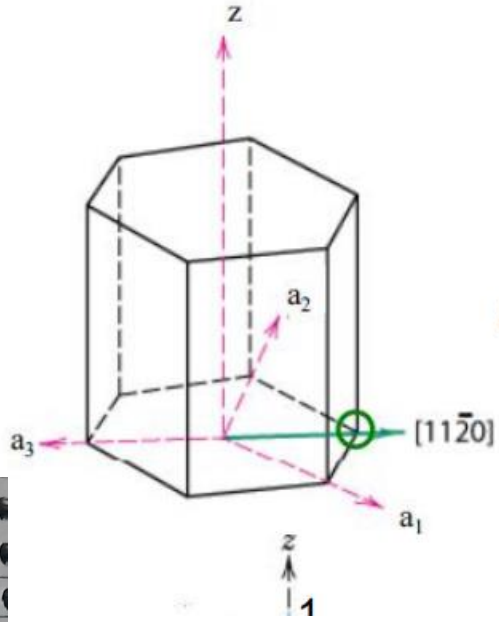
$$V = \frac{1}{3}(2v - u)$$

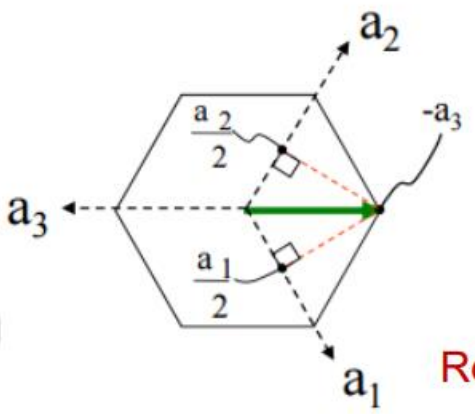
$$T = -(U+V)$$

$$W = w$$


X3
[1̄1̄0]
[010]
X2
120°
[100]
[110]
X1
X
X3
X2
X1
a3
a2
a1
X1[100] ⇒ [2/3 -1/3 -1/3 0] = [2 1̄ 1̄ 0]
X2[010] = [-1/3 2/3 -1/3 0] = [1̄ 2 1̄ 0]
X3[110] = [-1/3 -1/3 2/3 0] = [1̄ 1̄ 2 0]
<2 1̄ 1̄ 0> = [2 1̄ 1̄ 0] [1̄ 2 1̄ 0] [1̄ 1̄ 2 0]
U = 1/3 (2u − v)
V = 1/3 (2v − u)
T = -(U+V)
W = w

From slides





Projections

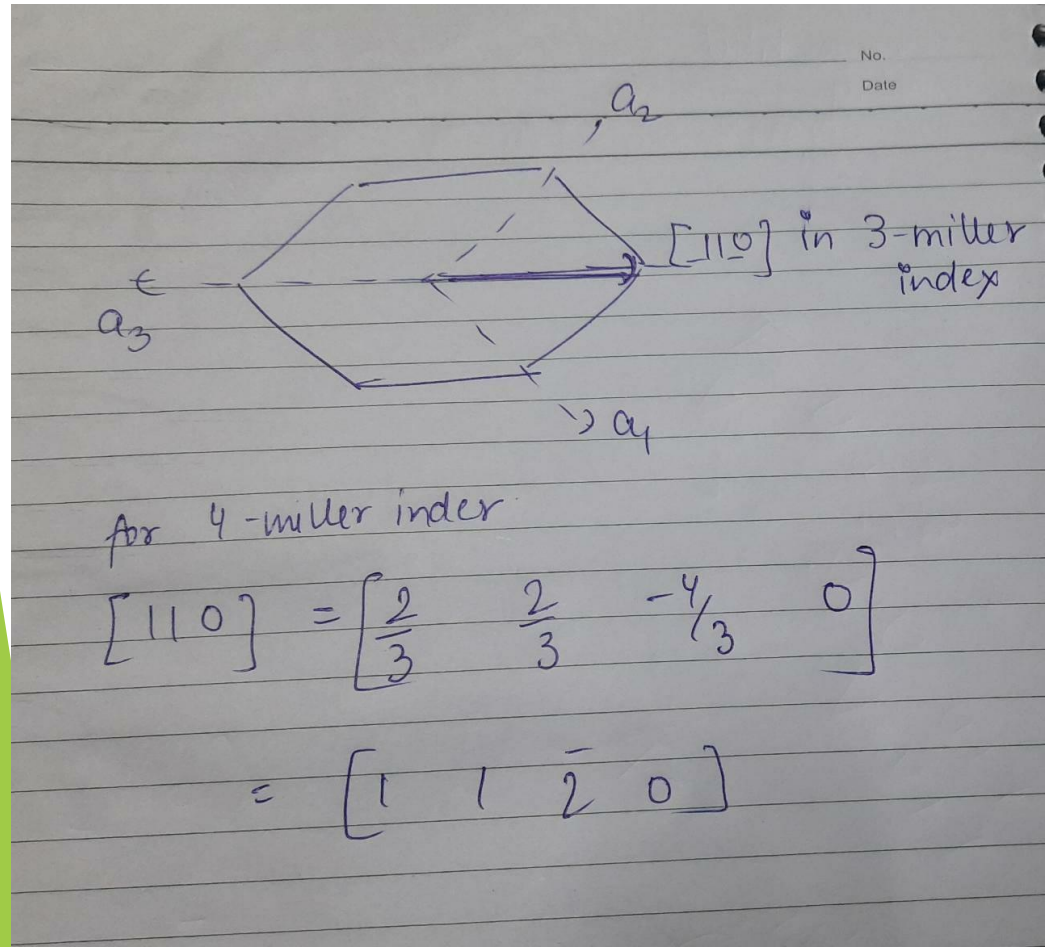
$a_1/2,\ a_2/2,\ -a_3,\ 0$

In terms of axis

1/2, 1/2, -1,0

Reduced to smallest integer

$[11\bar{2}0]$

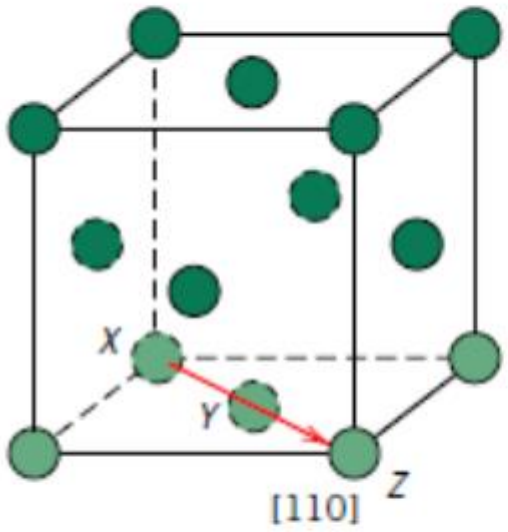


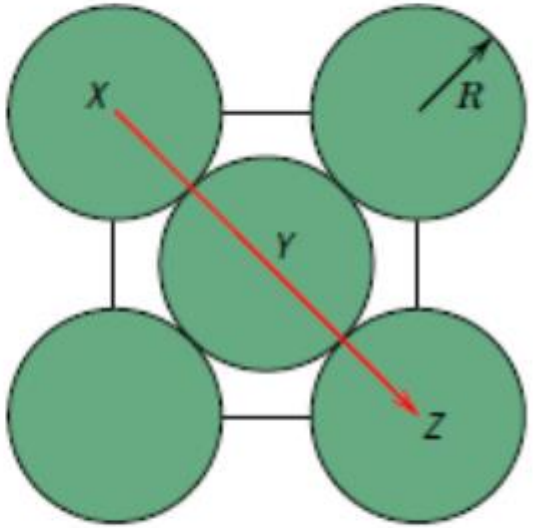
Density

- Linear
  - Direction
- Planar
  - Planes

Linear Density

Linear Density "LD" is defined as the number of atoms per unit length along a crystallographic direction.





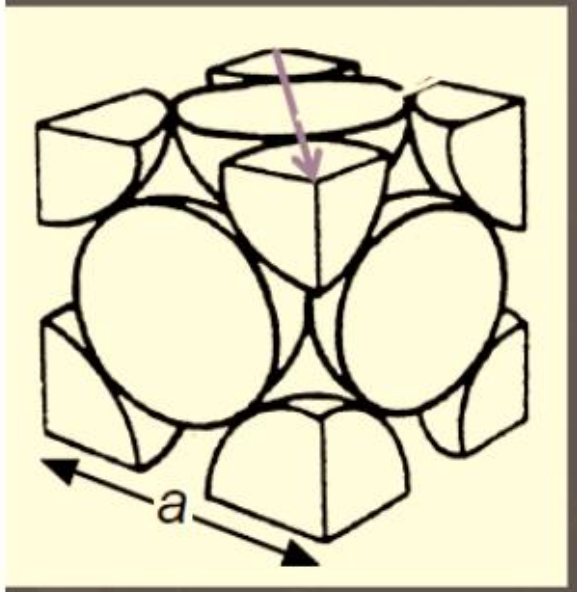
$$LD = \frac{Number\ of\ atoms\ centered\ on\ the\ direction\ vector}{length\ of\ the\ direction\ vector}$$

$$LD = \frac{2}{4R} = \frac{1}{2R}$$

Source: Prof Grewal's Slides

Example from slides

- Linear Density of Atoms ≡ LD = $\frac{\text{Number of atoms}}{\text{Unit length of direction vector}}$



ex: linear density of Al in [110] direction

a = 0.405 nm

atoms

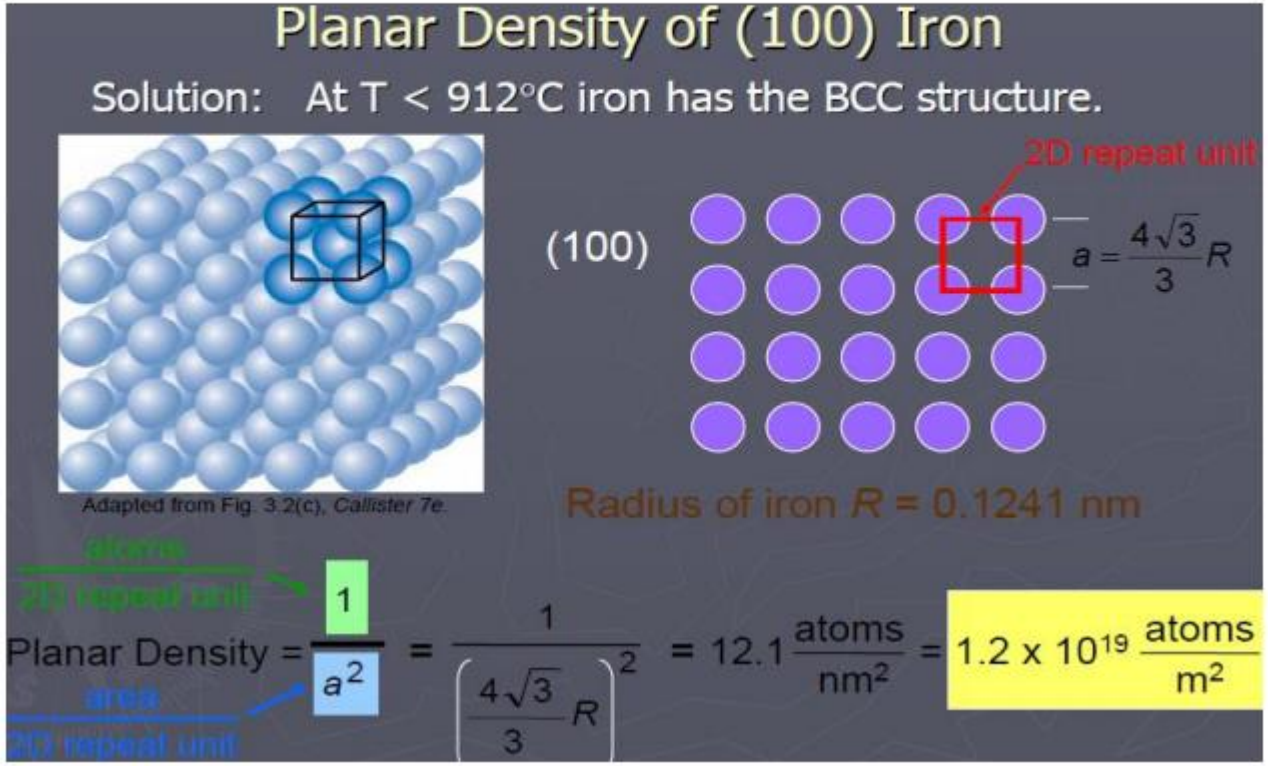
$$LD = \frac{2}{\sqrt{2}a} = 3.5\ \text{nm}^{-1}$$

length

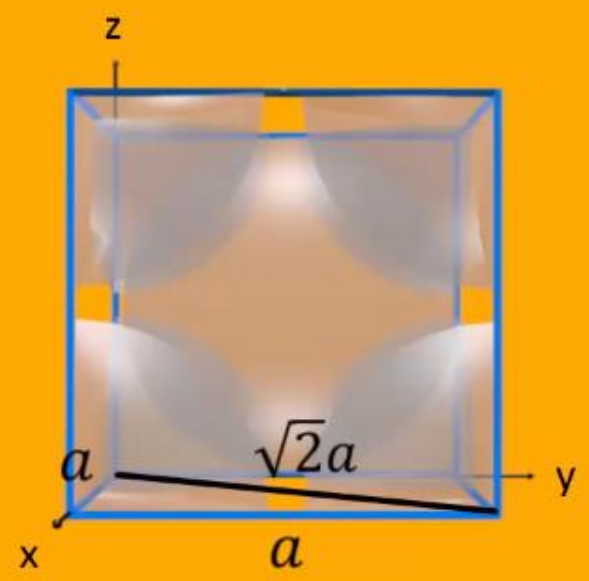
Planar Density

- Planar Density is defined as the number of atoms per unit area that are centered on a given crystallographic plane

$$PD = \frac{No.\,of\ atoms\ centered\ on\ the\ plane}{Area\ of\ the\ plane}$$



Explanation for relation b/w a and r

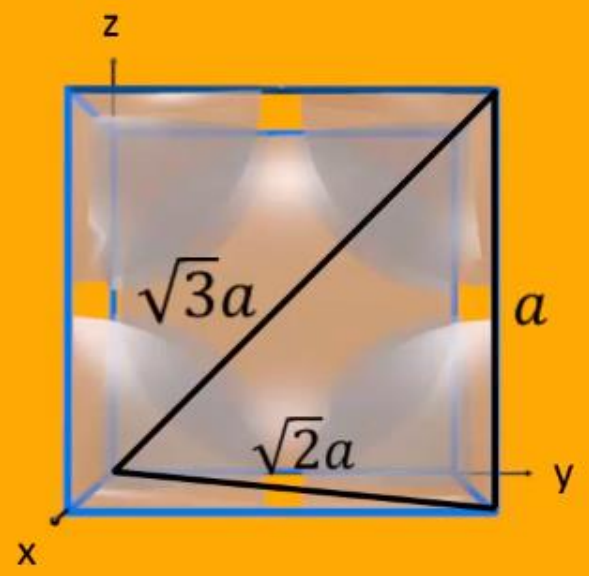


a = unit cell dimension
r = radius of atom

Using Pythagoras' Theorem:

$$c^2 = a^2 + b^2$$

$$hypotenuse^2 = a^2 + a^2$$
$$hypotenuse^2 = 2a^2$$
$$hypotenuse = \sqrt{2}a$$



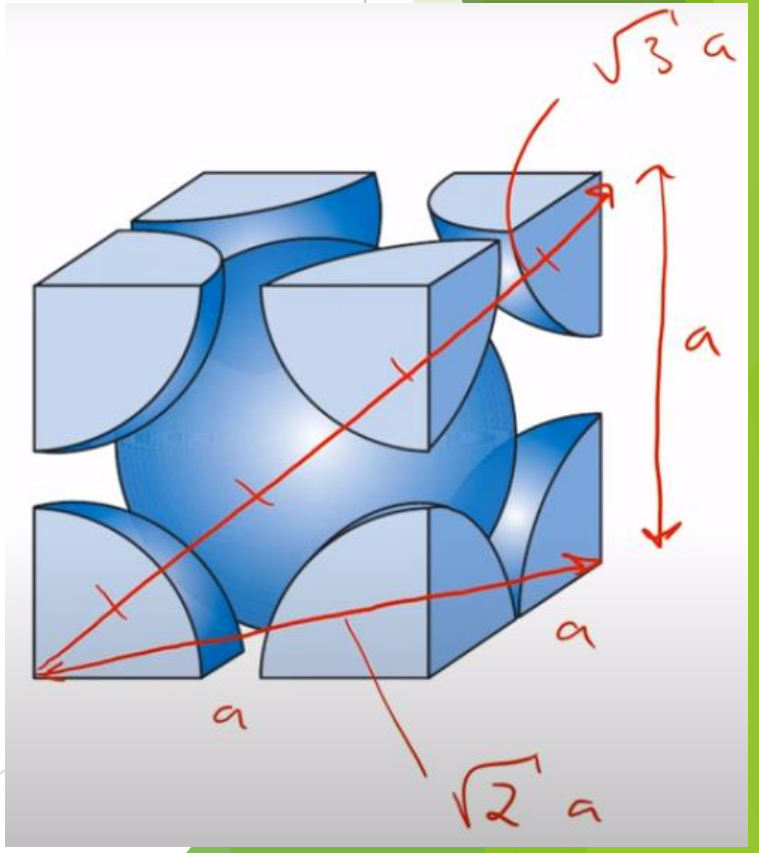
a = unit cell dimension
r = radius of atom

Using Pythagoras' Theorem:

$$c^2 = a^2 + b^2$$

$$hypotenuse^2 = a^2 + (\sqrt{2}a)^2$$
$$hypotenuse^2 = 3a^2$$
$$hypotenuse = \sqrt{3}a$$

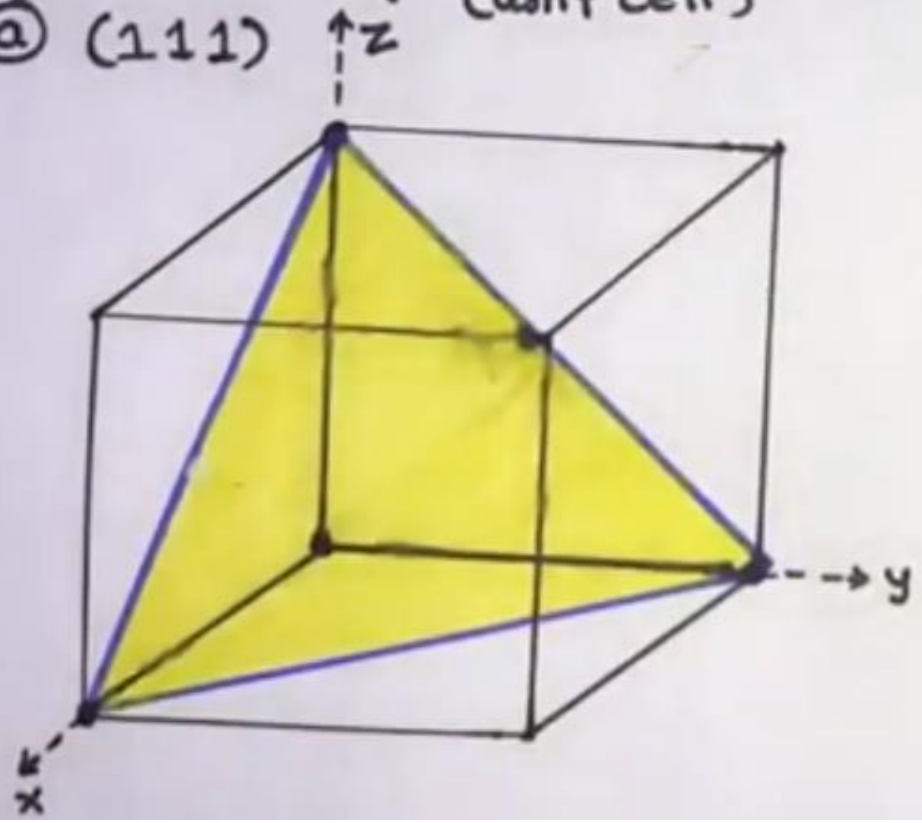
$$hypotenuse = \sqrt{3}a = 4r$$
$$\sqrt{3}a = 4r$$
$$\boxed{a = \frac{4}{\sqrt{3}}r}$$

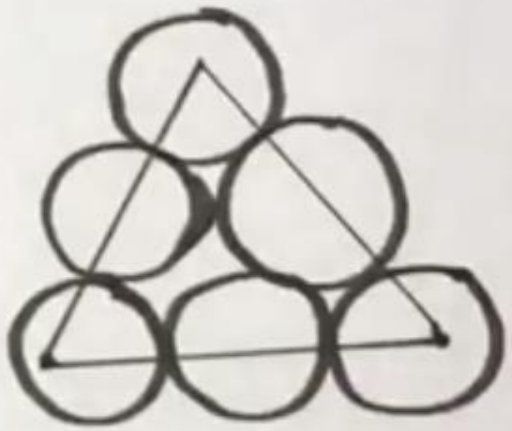


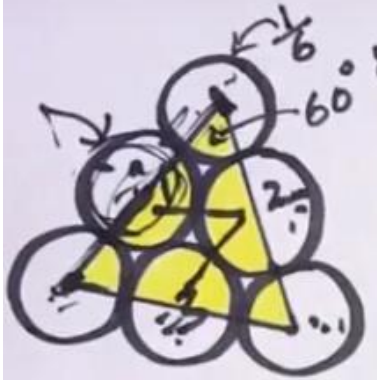
(111) FCC

1. Qn Find planar density of the following planes in FCC crystal structure (unit cell)

ⓐ (111)







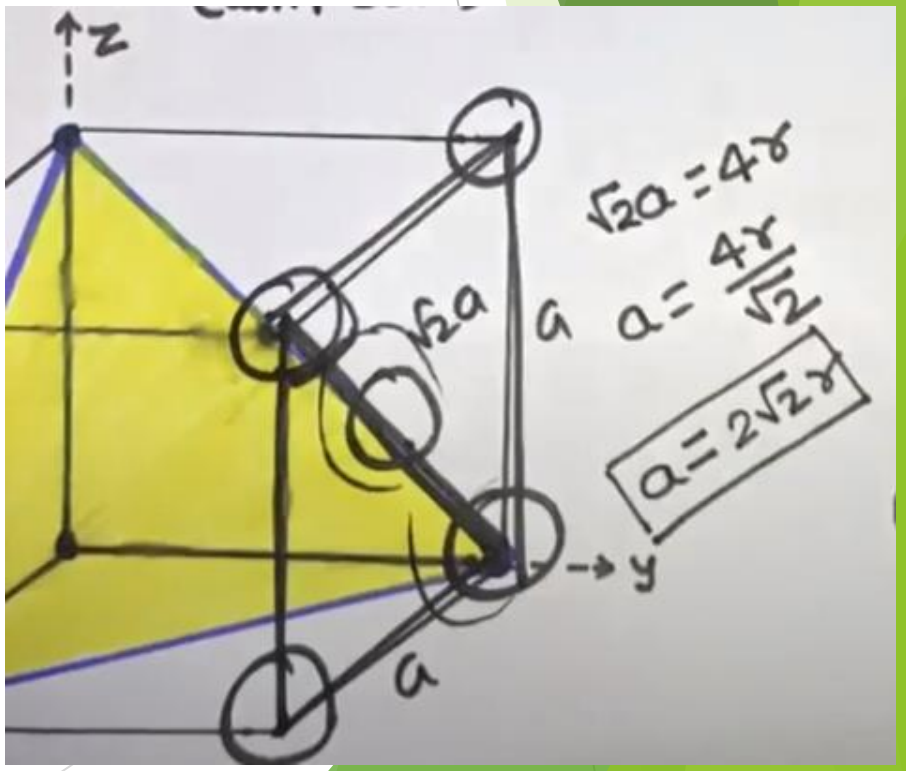
Effective no of atoms in the given plane

$$= \frac{1}{6} \times 3 + \frac{1}{2} \times 3$$

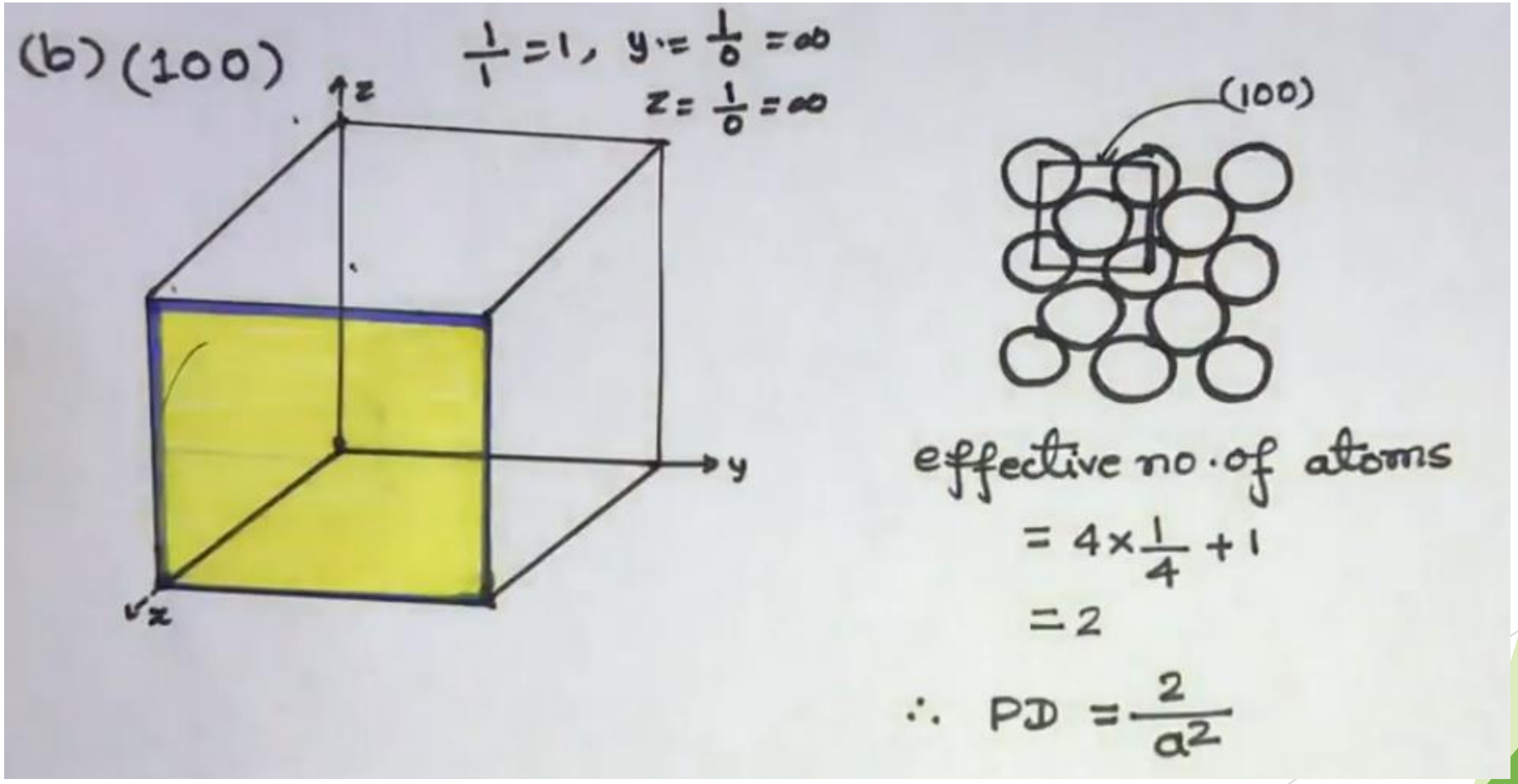
$$= 2$$

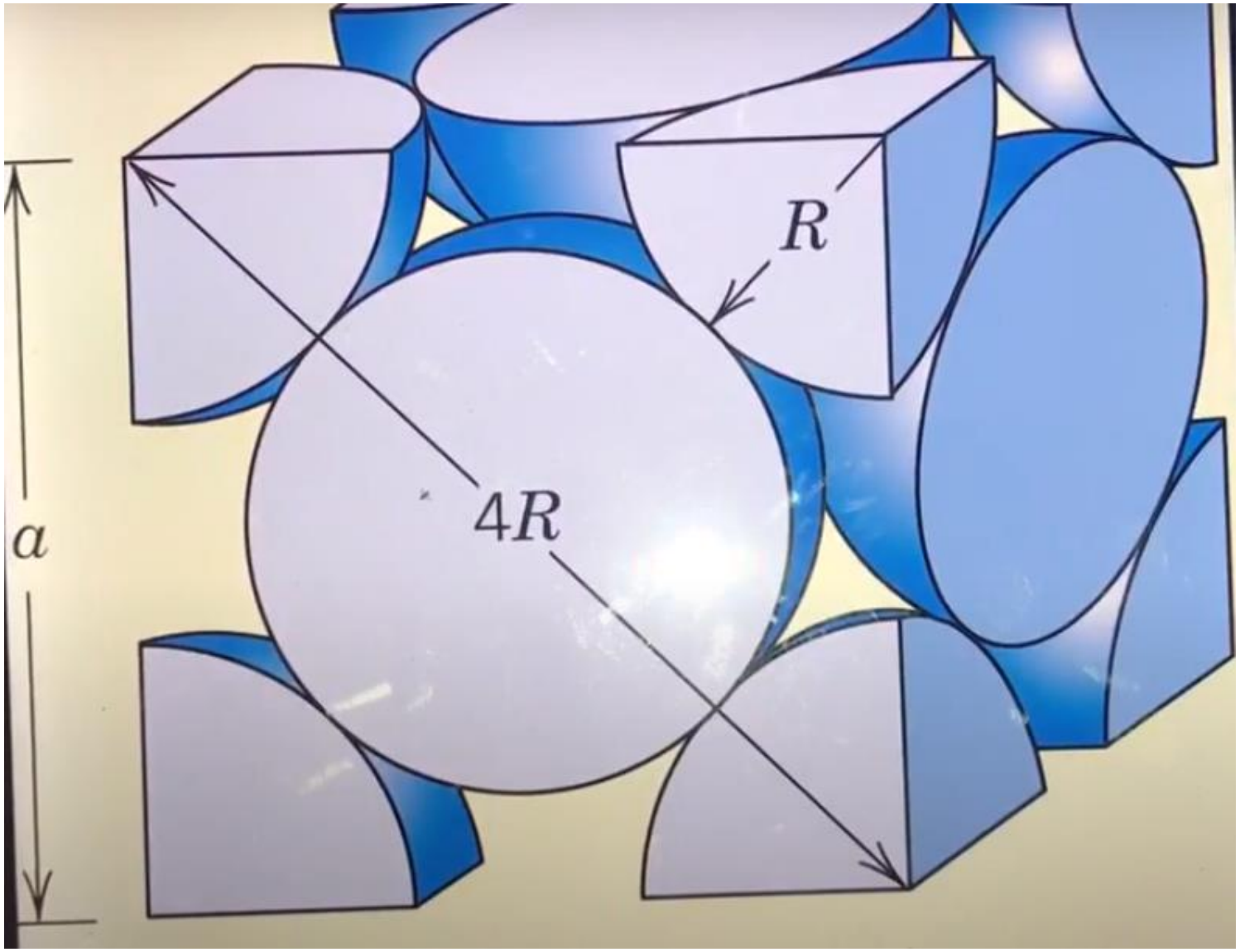
$$\therefore \quad PD = \frac{2}{\frac{\sqrt{3}}{4}(\sqrt{2}a)^2}$$

$$= \frac{2 \times 4}{\sqrt{3} \times 2a^2} = \frac{4}{\sqrt{3}a^2}$$

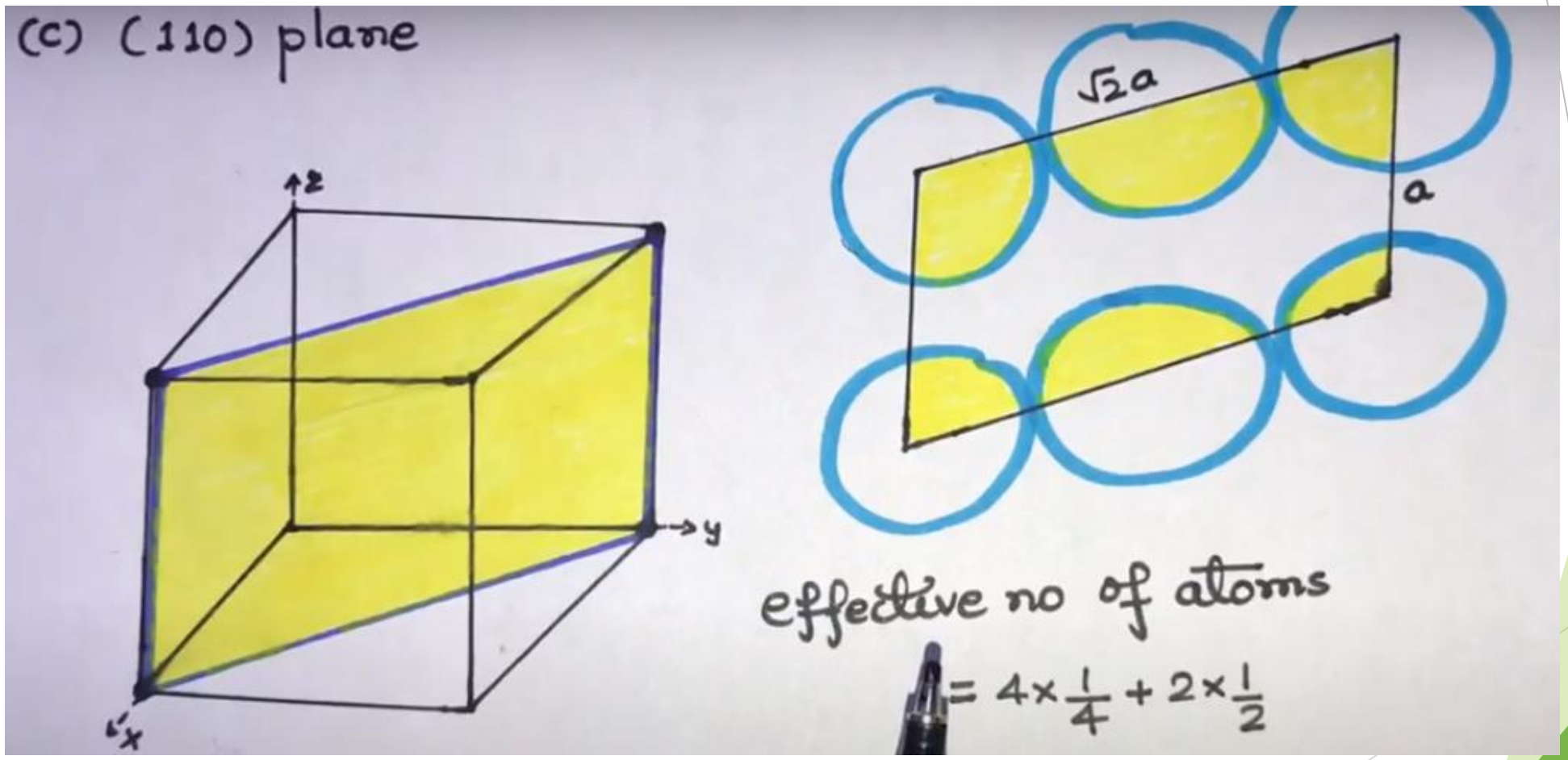


(110) FCC


(b) (100)
$\frac{1}{1} = 1$, $y = \frac{1}{0} = \infty$
$z = \frac{1}{0} = \infty$
z
y
x
(100)
effective no. of atoms
$= 4 \times \frac{1}{4} + 1$
$= 2$
$\therefore$ PD $= \frac{2}{a^2}$


R
4R
a

(110) FCC

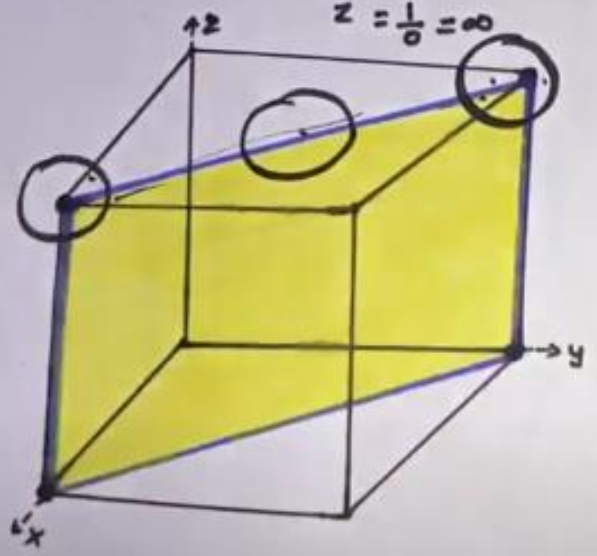


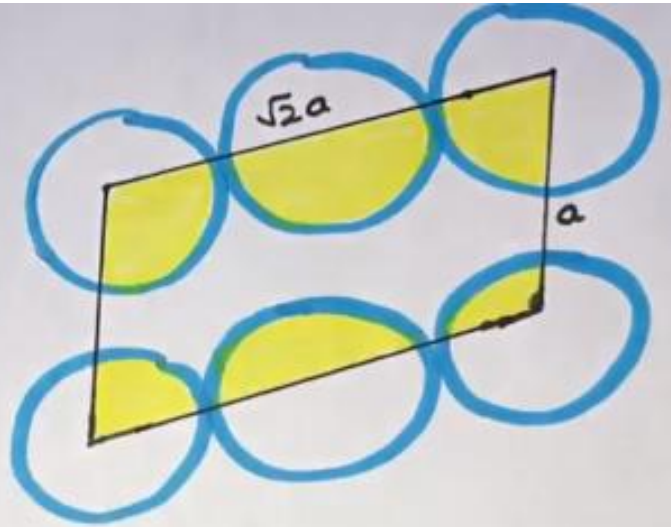
(c) (110) plane

$x = \frac{1}{1} = 1$

$y = \frac{1}{1} = 1$

$z = \frac{1}{0} = \infty$





effective no of atoms

$= 4 \times \frac{1}{4} + 2 \times \frac{1}{2}$

$= 2$

$\therefore \; PD = \frac{2}{\sqrt{2}a \cdot a} = \frac{2}{\sqrt{2}a^2} = \frac{\sqrt{2}}{a^2}$

$4r = \sqrt{2}\,a$

replacing in PD

$$PD = \frac{2}{4r \cdot \frac{4r}{\sqrt{2}}}$$

$$= \frac{2}{8\,16r^2\,\frac{\sqrt{2}}{2}}$$

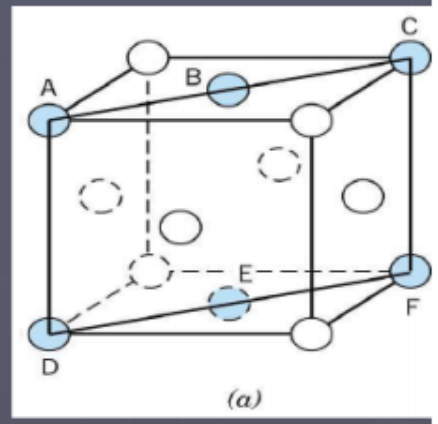
$$= \frac{2}{8\sqrt{2}\,r^2}$$

$$= \frac{1}{4\sqrt{2}\,r^2}$$

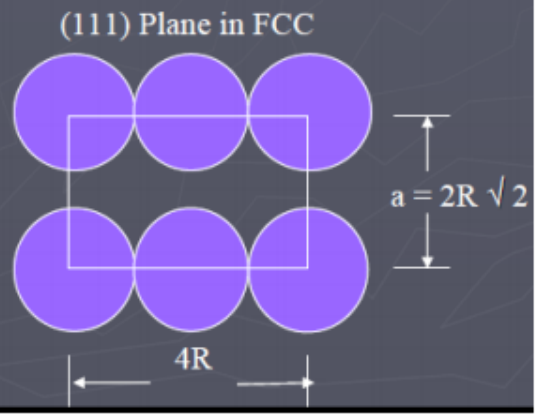
Planar Density of (110) plane in FCC

of atoms centered on the plane (110)
= 4(1/4) + 2(1/2) = 2 atoms

Area of the plane
= (4R)(2R √ 2) = 8R²2



$$PD_{110} = \frac{2\ atoms}{8R^2\sqrt{2}} = \frac{1}{4R^2\sqrt{2}}$$



What is the planar density of (111) plane in FCC? $\frac{1}{2\sqrt{3}R^2}$